ANSWERS DAY 3

SPOT 1

1. From 93 mmHg to 37 mmHg
2. Both flow and pressure will decrease
3. Largest in systemic veins (54%), 2nd largest in Pulmonary system (18%)

SPOT 2

1. U wave
2. Hypokalemia
3. Mod. Hypokalemia → Various arrhythmias; Sev. Hypokalemia → bradycardia and cardiac arrest

SPOT 3

1. Phase 0: Na+ enters; Phase 1: K+ exits; Phase 2: Ca++ enters and K+ exits; Phase 3: K+ exits; Phase 4: not part of action potential. It is resting membrane potential.
2. QRS complex
3. Passive

SPOT 4

1. A = Plasma membrane of adjacent cardiac muscle; B = Desmosome ; C = Gap junction; D = Action potential; E = Intercalated Disc
2. B = Cell adhesion (Hold cardiac cells together during contraction.); C = Electrical propagation
3. – 90 mV approx.

SPOT 5

1. A and D
2. C – It is equal to 130mL (in some books it is 120mL)
3. Stroke volume

SPOT 6

1. Third degree heart block
2. 40 / min
3. Because of vagal influence on SA node that keeps the heart rate low

SPOT 7

1. Cardiac conducting cell

ANS acts on phase 4 (prepotential or pacemaker potential)

2. Sympathetic → cause increased entry of Na+ and Ca++ inside conducting cell → cell reaches threshold early so HR increased
3. Parasympathetic → cause increase efflux of K+ ions → hyperpolarizing the cell → takes more time to reach threshold so decrease HR